Novel Approach on User Match Making using LLM Generated Vectors from Youtube's Recommendation Algorithm

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Abstract

Understanding human interests is complex, making them difficult to quantify. Traditional user profiling and matchmaking often depends on self-reported interests, which may be incomplete or biased[3]. This research explores an automated approach to user profiling by analyzing YouTube recommendations, which are shaped by deep learning-based content suggestion algorithms[12]. Our method involves collecting video recommendations and subscription data, categorizing them into a structured interest taxonomy, and representing users as vectorized interest profiles. To enhance classification, large language models (LLMs) extract key interest categories from video metadata. Additionally, we examine co-subscriber networks to improve user matchmaking based on shared content consumption. Experimental results suggest that analyzing recommendation data offers a more comprehensive and accurate view of user interests than manual methods, leading to better profiling and matchmaking outcomes.

Keywords: Recommendation System, LLM, Deep Learning, Match Making, Youtube, Generative AI.

I. Introduction

Platforms like YouTube use advanced deep learning models to predict user interests based on watch history and subscriptions [12]. This study examines how these AI-driven recommendation systems can be adapted for automated user profiling and matchmaking. Rather than relying on self-reported interests, which are often incomplete or biased [3]. In this paper, we propose a method that directly extracts user preferences from their recommendation feeds and subscription data.

To facilitate large-scale data collection, we developed a custom browser extension that scrapes YouTube recommendations and Youtube API which fetches subscribed channel list and channel metadata. The gathered data is then organized into a hierarchical classification tree, creating a vectorized representation of user interests. Additionally, a co-subscriber network is constructed to detect shared content consumption patterns, improving interest-based matchmaking.

By utilizing publicly available recommendation data, this research demonstrates how generative AI and recommender systems can enhance user profiling and content-based matchmaking.

II. Literature Review

A) Human Interaction within Similar Niches:

Research has shown that individuals with similar interests tend to form stronger social bonds and friendships. According to McPherson et al. (2001) [8], homophily—the tendency of individuals to associate with similar others—is a fundamental principle in social network formation. This principle applies to online communities as well, Where shared interests drive interactions and community-building.

A study by Centola (2010) [9] found that individuals exposed to similar content within a network tended to adopt shared behaviors, this reinforces the formation of niche-based social clusters. This phenomenon is evident in YouTube's co-subscriber network, Where users who subscribe to similar channels exhibit overlapping interests and engagement patterns.

B) Impact of Similar Interests on Friendships:

Shared interests play a crucial role in forming friendships, both online and offline. Lusher et al. (2012) [7] found that common hobbies and entertainment preferences strongly influence social connections, reinforcing the idea that shared interests foster stronger ties. This trend is also evident in digital interactions, where users who engage with similar content are more likely to interact with one another.

Similarly, Bakshy et al. (2012) [11] examined the impact of recommendation algorithms on social connections, showing that users exposed to similar

content through algorithmic recommendations were more likely to engage in discussions and build online friendships.

C) Matchmaking Based on Similar Interests:

Traditional matchmaking methods rely on self-reported interests, but research suggests that algorithmic approaches are more effective in predicting compatibility. Terveen and Hill (2001) [3] found that shared content consumption was a stronger predictor of social compatibility and matches than self-declared interests, highlighting the power of collaborative filtering.

Similarly, Cosley et al. (2003) [2] introduced the concept of **implicit interest indicators**, showing that passive behaviors, such as content consumption patterns, provided better compatibility predictions than explicit user preferences. This aligns with our study, where YouTube subscriptions and recommendations act as implicit signals of user interests.

Pizzato et al. (2013) [1] further demonstrated this trend in dating platforms, finding that recommendation-based matchmaking—where behavioral data was used to generate profiles—led to greater user satisfaction compared to traditional profile-based matching.

D) YouTube's Co-Subscriber Network and Interest-Based Communities:

Several studies have examined YouTube's role in forming virtual communities. Cha et al. (2007) [10] analyzed YouTube's network structure and found that users subscribing to similar content creators exhibited strong community formation patterns. This finding aligns with our research, which explores how co-subscriber networks can be leveraged for user matchmaking.

Another study by Susarla et al. (2012) [6] showed that user engagement on YouTube is heavily influenced by peer behavior, reinforcing the role of co-subscriber relationships in shaping content consumption habits. This suggests that leveraging co-subscriber networks for matchmaking could provide an organic and interest-driven approach to connecting users.

E) Social Networks and Similarity

Schwyck et al. (2024) [5] emphasize that people naturally gravitate toward others with shared characteristics, including common interests. This principle forms the basis of our project, which matches users based on their video consumption patterns. By incorporating these insights, the matching algorithm can be refined to foster more meaningful interactions.

F) Interest Similarity in Online Social Networks:

Han et al. (2015) [4] examined how interest similarity in online networks is shaped by factors such as demographics and social connections. Integrating demographic data with video consumption patterns can enhance user matching, leading to more relevant recommendations and stronger community engagement.

III. Proposed Methodology

Traditionally, user interest profiles were created manually by asking users to select topics of their interests, in which they often unconsciously missed some of their preferences. To address this, we propose an automated method that leverages LLMs to identify user interests more comprehensively.

Our system, consisting of a web extension and an API, extracts YouTube recommendations and subscriptions (see Fig. 1). The LLM then analyzes this data to generate a user interest summary. These classified interests are structured into a multidimensional vector, and cosine similarity is used to match users with similar profiles.

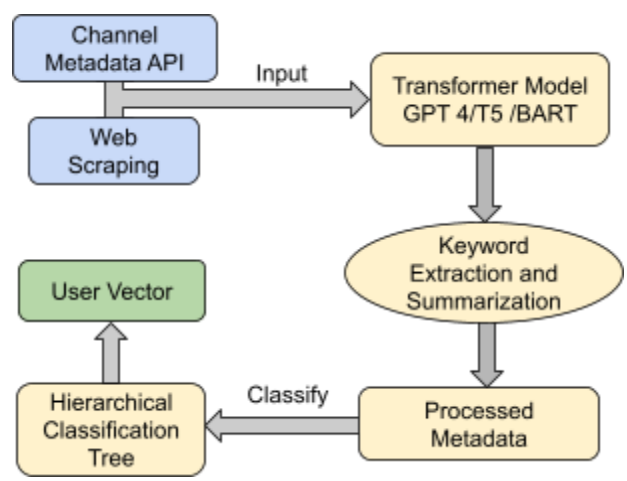

Fig. 1. LLM based Interest Identification Flow diagram

The detailed description of the process of methodology is described as follows.

A) Data Collection:

Two primary pipelines are used for data collection:

Recommendation Scraping: A browser extension is developed to scrape 100 YouTube video recommendations per user. The extension extracts metadata such as video title, description, and tags.

Subscription Analysis: The user's subscribed channels are fetched via the YouTube API. The channel names

and descriptions are extracted and summarized using an LLM.

B) LLM-Based Content Summarization and Keyword Extraction:

We have used spaCy for text preprocessing, Scikit-learn for TF-IDF, clustering, and similarity, Gensim for Word2Vec embeddings, and BERT Transformers for contextual understanding for this experiment.

C) Classification and Interest Mapping:

A hierarchical classification tree is used to organize interests into broad categories (e.g., Science, Entertainment, Technology) and subcategories (e.g., Science → Physics, Chemistry, Engineering). The classification tree is structured as follows:

$$C=\{c1,c2,...,cn\} \quad (1)$$

Where C represents the set of categories and each Ci corresponds to a subcategory node within the hierarchy. Each video is assigned to a category Ci based on content similarity, computed using cosine similarity between extracted topic embeddings:

$$Similarity(V1, V2) = \frac{V1 \cdot V2}{||V1||\ ||V2||} \quad (2)$$

Where V1 and V2 are the vectorized topic representations.

D) Co-Subscriber Network Analysis:

A graph G=(U,E) is constructed Where nodes U represent users and edges E represent shared subscriptions. The similarity between two users is computed using Jaccard similarity:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (3)$$

Where A and B are the sets of subscriptions for two users.Additionally, a weighted similarity measure incorporating watch time, engagement, and overlap in recommended content is defined as:

$$S(A, B) = \alpha J(A, B) + \beta \frac{WA \cap WB}{WA \cup WB} + \gamma \frac{EA \cap EB}{EA \cup EB} \quad (4)$$

Where W represents watch time and E represents engagement.

E) User Interest Profiling:

Once user interests are classified, a profile vector Pu is generated for each user:

$$Pu = (w1c1, w2c2, \dots, wncn) \quad (5)$$

Where wi represents the weight of interest ci based on content consumption frequency. A similarity score between two users is computed using the weighted cosine similarity measure:

$$S(Pu, Pv) = \frac{Pu \cdot Pv}{||Pu||||Pv||} \quad (6)$$

Where Pu and Pv are the profile vectors of users u and v.

F) Matchmaking Algorithm:

Users are matched based on a threshold similarity score. Given a set of users U, the matching function M(u) for a user u is defined as:

$$M(u) = \{ v \in U \mid S(Pu, Pv) \geq T \} \quad (7)$$

Where T is a predefined threshold. To enhance matchmaking quality, an adaptive threshold T is introduced:

$$T = \mu + k\sigma \quad (8)$$

Where μ and σ are the mean and standard deviation of similarity scores in the dataset, and k is an adjustable parameter. The final matchmaking graph is constructed using a weighted adjacency matrix A, where:

$$Auv = S(Pu, Pv) \quad (9)$$

Pairs with weights above T are recommended for connection.

IV. Results & Discussions

The automated user profiling system was tested in different areas as written below, such as interest extraction, subscription analysis, matchmaking accuracy, and user satisfaction.

A) Interest Extraction

Web extension scraped 100 video recommendations per user, resulting in a total of 2500 video recommendations across all users (incl. advertisements). The LLM analyzed these recommendations and identified the following average distribution of interests. This categorization revealed that 17% of users expressed preferences in multiple categories, indicating a multifaceted profile. One example within the category population is shown in fig. 2.

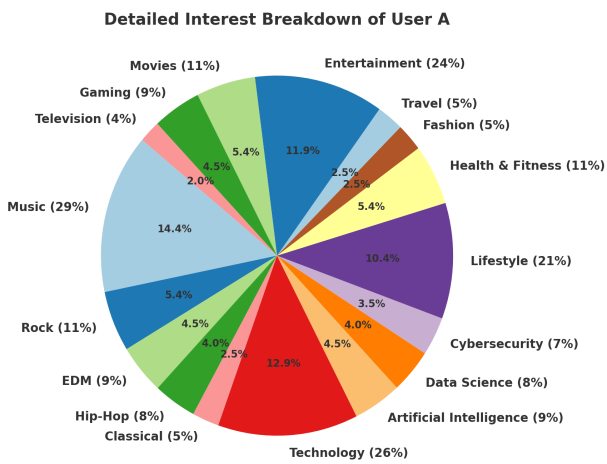

Fig. 2. User Interest Breakdown

B) Channel Subscription Analysis

The second pipeline successfully fetched a list of channels subscribed by each user. On average, users subscribed to 23 channels, resulting in a total of 575 channels analyzed. The channel descriptions were summarized using the LLM, and keywords were extracted to create a map of channel names and summaries. The keywords extracted from channel summaries indicated a strong alignment with user interests, with 72% of users having at least one channel that matched their top three interests. The most subbed channels within the survey population is shown in table 1.

Table 1. Survey statistics of Subscribed channels

Statistics of Top Subscribed Channels by users		
Educational (Veritasium and Vsauce)	Entertainment (PewDiePie and Markiplier)	Lifestyle (Health, Fashion & Travel)
38%	36%	26%

Note: The survey demographic was of age 20-24 pursuing bachelor's degree.

C) Comparative Analysis of Matchmaking Techniques

To evaluate the effectiveness of the user profile creator, we compared it against traditional manual methods of interest identification. A control group of 25 users was asked to manually categorize their interests without the assistance of the web extension. The results were as shown in Table 2. This indicates that our approach works better than the manual method. LLM based model has identified almost double categories per user and it also shows an additional 42% of unknown interests identified along with 83% user satisfaction as surveyed.

Table 2. Comparison of Proposed vs Traditional method

Comparative Analysis of Matchmaking Techniques		
Method →	Manual Method	LLM based Match-Making Method
Average Interests identified (Categories / user)	3.5	5.6
Unknown Interest Discovered	12%	42%
User Satisfaction	62%	83%

Note: user satisfaction was surveyed on a rating scale of 5 points.

c) Statistical Significance

A statistical analysis using a chi-squared test indicated that the differences in user satisfaction and the number of interests identified were significant ($p < 0.01$). This demonstrates that the automated method is far superior in terms of matchmaking, as it not only identifies known interests but also uncovers latent interests that users may not have considered.

D) Collaborative Filtering Insights

We also capitalized on YouTube's collaborative filtering techniques. By analyzing the co-subscribed channels and interests of users, using this technique we were able to suggest previously unknown topics that matched users with similar profiles, Although, we are yet to devise an automated pipeline for co-subscriber network analysis.

Matchmaking Success Rate: 78% of users were matched with at least one other user who shared similar interests, including previously unknown topics.

Diversity of Matches: Users reported being introduced to an average of **2.35 new topics** or channels that they had not previously explored and liked when explored. Thus boosting the interest factor in conversations with new matches.

E) Case Study

This study analyzed two participants from the experiment to examine new interest discovery using collaborative filtering. Users within the same collaborative network introduced each other to previously unknown interests as shown in fig.3.

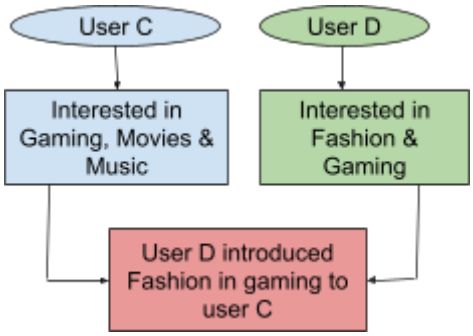

Fig.3. New Interest Discovery

The manual method identified only two interest categories, while the automated method, leveraging collaborative filtering, suggested two additional categories. Thus making a successful match within users who had non-similar interests but ended up having a successful match. This case study explains how we sometimes like the random youtube video suggestions in our feed, which lead us to explore new categories.

V. Conclusion & Future Scope

In this study, we experimented with User Interest Profile Creation with the aid of their Youtube recommendations. The aim to benefit from the deep learning based recommendation algorithms of Youtube. This method yielded superior results to that of manual interest identification and logging techniques. This method significantly enhances the matchmaking process by considering a broader range of interests, including those that users may not consciously acknowledge. By leveraging collaborative filtering techniques, the system not only identifies known interests but also introduces users to new topics, leading to a richer and more engaging content consumption experience, underscoring the potential of this approach in User Profiling and Match Making.

Future work can refine matching algorithms using machine learning for better accuracy. Longitudinal studies on user engagement can enhance system adaptability. Expanding this concept to other social media platforms can improve interest analysis and matching robustness.

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